DATE VALIDATION

Aim:

To build a system using COBOL that checks whether a date entered by the user is valid in YYYYMMDD format.

Algorithm:

Step 1: Start the program.

Step 2: Accept the date from the user in YYYYMMDD format.

Step 3: Check that the input contains exactly eight numeric characters.

Step 4: Separate the input into year (YYYY), month (MM), and day (DD).

Step 5: Check that the year is greater than zero and the month is between 1 and 12.

Step 6: Determine whether the year is a leap year. A year divisible by 400 is a leap year; a year divisible by 100 but not 400 is not; otherwise, a year divisible by 4 is a leap year.

Step 7: Determine the maximum number of days in the selected month.

Step 8: Check whether the day is between 1 and the maximum number of days allowed for that month.

Step 9: Display VALID DATE INPUT if all checks pass; otherwise, display INVALID DATE INPUT.

Step 10: Stop the program.

Code:

```
IDENTIFICATION DIVISION.
PROGRAM-ID. DATE-VALIDATION.

DATA DIVISION.
WORKING-STORAGE SECTION.
    01 DATE-INPUT PIC X(8).
    01 YYYY       PIC 9(4).
    01 MM         PIC 99.
    01 DD         PIC 99.
    01 LEAP       PIC X VALUE "N".
    01 MAX-DAY    PIC 99 VALUE 0.
    01 VALID-FLAG PIC X VALUE "N".

PROCEDURE DIVISION.
    DISPLAY "ENTER DATE IN YYYYMMDD FORMAT:"
    ACCEPT DATE-INPUT

    IF DATE-INPUT IS NUMERIC
        MOVE DATE-INPUT(1:4) TO YYYY
        MOVE DATE-INPUT(5:2) TO MM
        MOVE DATE-INPUT(7:2) TO DD

        MOVE "N" TO LEAP
        MOVE 0 TO MAX-DAY

        IF YYYY > 0
            IF FUNCTION MOD(YYYY, 400) = 0
                MOVE "Y" TO LEAP
            ELSE
                IF FUNCTION MOD(YYYY, 100) = 0
                    MOVE "N" TO LEAP
                ELSE
                    IF FUNCTION MOD(YYYY, 4) = 0
                        MOVE "Y" TO LEAP
                    END-IF
                END-IF
            END-IF

            EVALUATE MM
                WHEN 1
                WHEN 3
                WHEN 5
                WHEN 7
                WHEN 8
                WHEN 10
                WHEN 12
                    MOVE 31 TO MAX-DAY
                WHEN 4
                WHEN 6
                WHEN 9
                WHEN 11
                    MOVE 30 TO MAX-DAY
                WHEN 2
                    IF LEAP = "Y"
                        MOVE 29 TO MAX-DAY
                    ELSE
                        MOVE 28 TO MAX-DAY
                    END-IF
                WHEN OTHER
                    MOVE 0 TO MAX-DAY
            END-EVALUATE

            IF MM >= 1 AND MM <= 12
                IF DD >= 1 AND DD <= MAX-DAY
                    MOVE "Y" TO VALID-FLAG
                END-IF
            END-IF
        END-IF
    END-IF

    IF VALID-FLAG = "Y"
```

```
    DISPLAY "VALID DATE INPUT"
ELSE
    DISPLAY "INVALID DATE INPUT"
END-IF

STOP RUN.
```

Output:

Example 1: Valid date

```
ENTER DATE IN YYYYMMDD FORMAT:
20240229
VALID DATE INPUT
```

Example 2: Invalid date (April has only 30 days)

```
ENTER DATE IN YYYYMMDD FORMAT:
20240431
INVALID DATE INPUT
```

Example 3: Invalid date (1900 is not a leap year)

```
ENTER DATE IN YYYYMMDD FORMAT:
19000229
INVALID DATE INPUT
```

Result:

Thus, a date validation system using COBOL is designed to check the input format, month, day, and leap-year rules, and display the expected result.